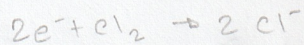
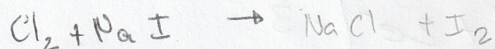
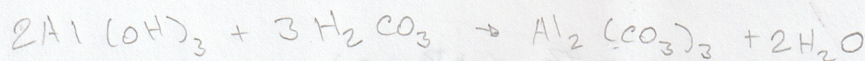
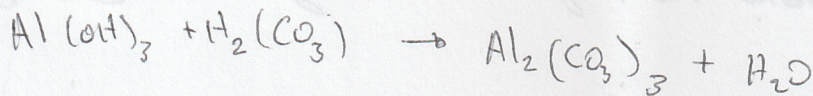
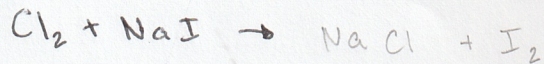
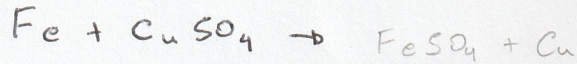


1) Completely balance:

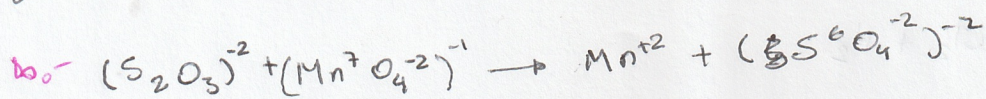


$$30\text{A} \rightarrow 30\text{C/s}$$

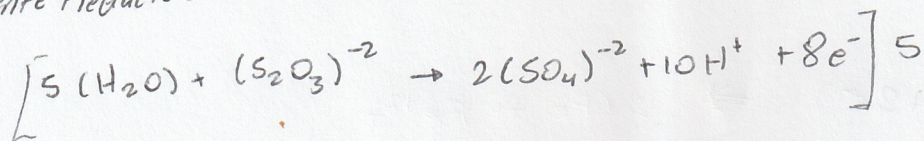
Injection 30 A

$$\begin{aligned} 10\text{ L Cl}_2 &\times \frac{0.71\text{g Cl}_2}{1\text{ L Cl}_2} \times \frac{1\text{ mol Cl}_2}{70.9\text{g Cl}_2} \times \frac{2\text{ mol } e^-}{1\text{ mol Cl}_2} \times \frac{96,500\text{C}}{1\text{ mol } e^-} \times \frac{1\text{sec}}{30\text{Coulombs}} = \boxed{644\text{sec}} \\ &= \boxed{10.73\text{ min}} \end{aligned}$$

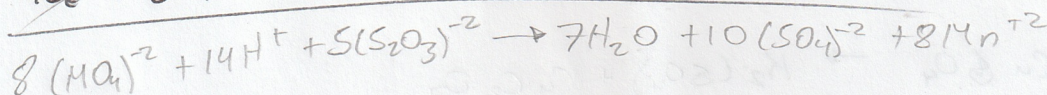
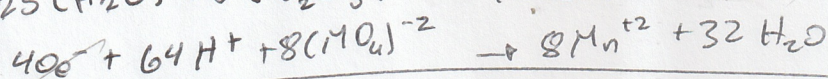
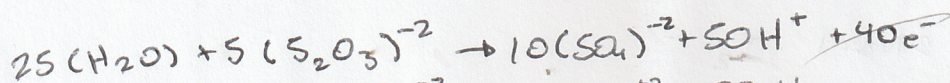
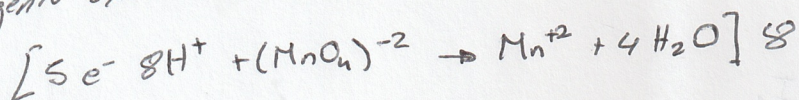
38.- Ajuste las siguientes reacciones redox que tienen lugar en medio ácido.



Agente Reductor:

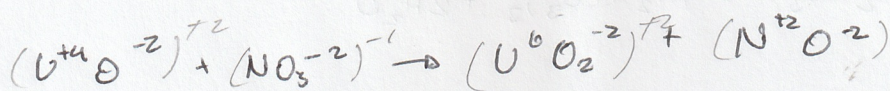


Agente oxidante:

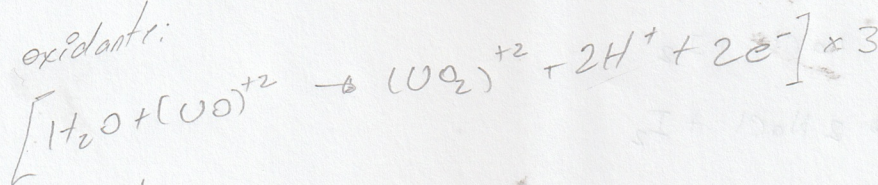


• Ajuste las siguientes reacciones redox que tienen lugar en medio ácido

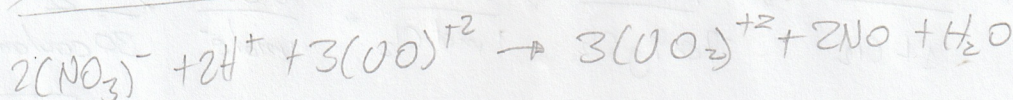
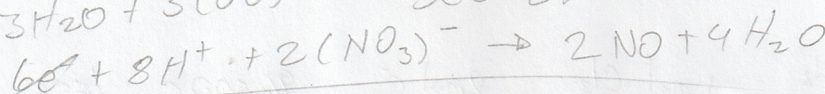
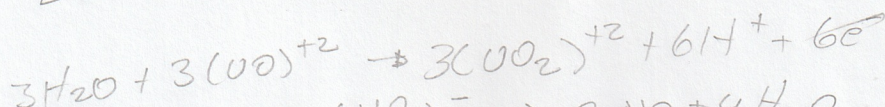
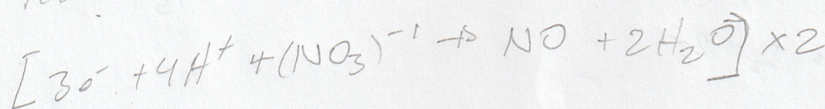
37.-
d)



Agente oxidante:



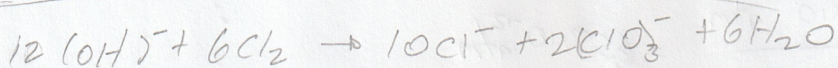
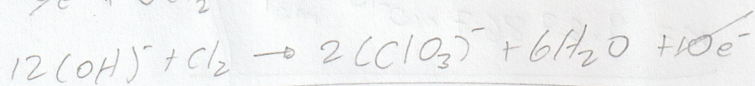
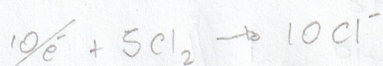
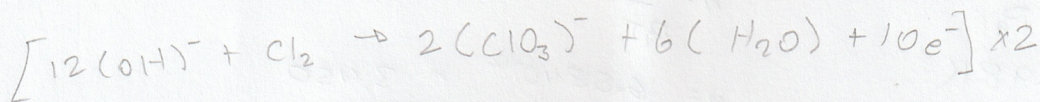
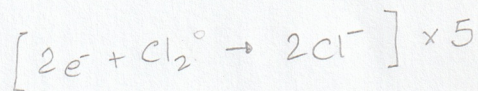
Agente reductor



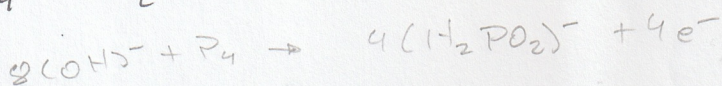
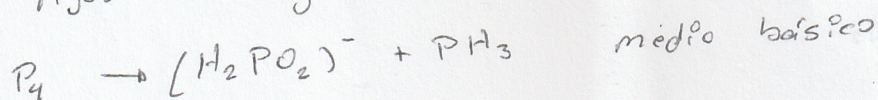
41.- Ajuste las siguientes ecuaciones de reacciones de desproporción:

Lester Iván Hernández H.
20242000637

a.- $\text{Cl}_2 \rightarrow \text{Cl} + [\text{ClO}_3]^-$ medio básico

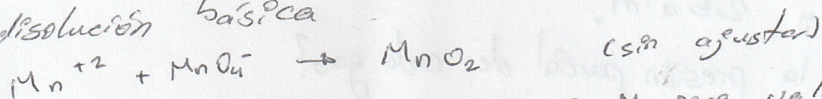


42.- Ajuste las siguientes ecuaciones de reacción de descomposición

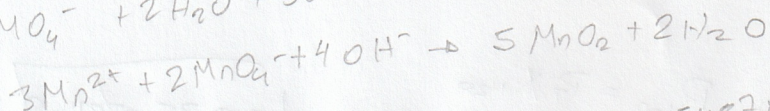
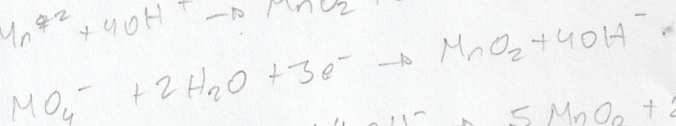
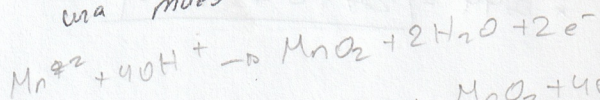


66.- La concentración de Mn^{+2} puede determinarse por valoración con

MnO_4^- en disolución básica



Se necesitan 37.21 ml de KMnO_4 (aq) 0.04162 M para valorar una muestra de 25.00 ml de Mn^{+2} ¿Cuál es la $[\text{Mn}^{+2}]$ en la muestra?



$$\text{moles } \text{Mn}^{+2} = \frac{3}{2} \text{ moles } \text{MnO}_4^- = 1.5 \times 1.5487 \times 10^{-3} \text{ mol } \text{MnO}_4^- = 2.3230 \times 10^{-3} \text{ moles } \text{Mn}^{+2}$$

$$[\text{Mn}^{+2}] = \frac{2.3230 \times 10^{-3} \text{ mol}}{0.025 \text{ L}} = 0.0929 \text{ M}$$

6.5.-

6.- ¿Cuántas moléculas de N_2 quedan en un sistema de ultra-alto vacío con un volumen de 3.45 m^3 cuando la presión se reduce a $6.67 \times 10^{-7} \text{ Pa}$ a 25°C ?

$$V = 3.45 \text{ m}^3$$

$$P = 6.67 \times 10^{-7} \text{ Pa}$$

$$T = 25^\circ\text{C} = 298 \text{ K}$$

$$R = 0.0821 \frac{\text{atm} \cdot \text{L}}{\text{mol} \cdot \text{K}}$$

$$PV = nRT$$

$$n = \frac{PV}{RT}$$

$$n = \frac{6.67 \times 10^{-7} \times 3.45}{0.0821 \times 298}$$

$$n = 9.27867 \times 10^{-10} \text{ mol}$$

$$6.67 \times 10^{-7} \text{ Pa} \times \frac{9.27 \times 10^{-6} \text{ atm}}{1 \text{ Pa}} = 6.58 \times 10^{-12} \text{ atm}$$

$$3.45 \text{ m}^3 \times \frac{1000 \text{ L}}{1 \text{ m}^3} = 3450 \text{ L}$$

6.12.-

6.- Una mezcla de 0.197 mol de CO_2 + 0.00278 mol de H_2O está a 30°C y 2.5 atm .

¿Cuál es la presión parcial de cada gas?

$$\frac{n_{\text{CO}_2}}{n_{\text{CO}_2} + n_{\text{H}_2\text{O}}} = X_{\text{CO}_2} \Rightarrow \frac{P_{\text{CO}_2}}{P_T} \Rightarrow P_{\text{CO}_2} = P_T X_{\text{CO}_2} = 2.5 \times 0.99 = \boxed{2.475 \text{ atm CO}_2}$$

$$\frac{0.197}{0.197 + 0.00278} = 0.99$$

$$\frac{n_{\text{H}_2\text{O}}}{n_{\text{H}_2\text{O}} + n_{\text{CO}_2}} = X_{\text{H}_2\text{O}} \Rightarrow \frac{P_{\text{H}_2\text{O}}}{P_T} \Rightarrow P_{\text{H}_2\text{O}} = P_T X_{\text{H}_2\text{O}} = 2.5 \times 0.014 = \boxed{0.035 \text{ atm H}_2\text{O}}$$

$$\frac{0.00278}{0.00278 + 0.197} = 0.014$$